

HDOC DAY-34

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BATCH:15

QUESTION:

Write a menu-driven C program using separate functions to verify:

1. Left distributive property:
$$\mathbf{A \times (B + C) = (A \times B) + (A \times C)}$$
2. Right distributive property:
$$\mathbf{(A + B) \times C = (A \times C) + (B \times C)}$$
3. Exit.

Aim:

To write a menu-driven C program using separate functions to verify the **left and right distributive properties of matrix multiplication** using variable-length 2D arrays.

Algorithm

1. Start.
2. Display the menu.
3. Read the user's choice.
4. If choice is **1**, input matrices A, B and C and verify:
$$\mathbf{A \times (B + C) = (A \times B) + (A \times C)}$$
5. If choice is **2**, input matrices A, B and C and verify:
$$\mathbf{(A + B) \times C = (A \times C) + (B \times C)}$$
6. Display both LHS and RHS matrices.
7. Compare LHS and RHS and display whether the property is verified.
8. If choice is **3**, exit the program.

9. Stop.

Functions Required

1. readMatrix() – To input matrix elements.
2. addMatrix() – To add two matrices.
3. multiplyMatrix() – To multiply two matrices.
4. displayMatrix() – To display a matrix.
5. compareMatrix() – To compare LHS and RHS matrices.

Test Case Choice Matrix Size Expected Result

1	1	2×2	LHS = RHS
2	2	2×2	LHS = RHS
3	3	—	Program exits

>C PROGRAM:

```
1#include <stdio.h>
2
3void read(int r,int c,int a[r][c]){
4    for(int i=0;i<r;i++){
5        for(int j=0;j<c;j++){
6            scanf("%d",&a[i][j]);
7        }
8    }
9
10 void add(int r,int c,int a[r][c],int b[r][c],int x[r][c]){
11     for(int i=0;i<r;i++){
12         for(int j=0;j<c;j++){
13             x[i][j]=a[i][j]+b[i][j];
14         }
15     }
16
17 void mul(int r1,int c1,int c2,int a[r1][c1],
18          int b[c1][c2],int x[r1][c2]){
19     for(int i=0;i<r1;i++){
20         for(int j=0;j<c2;j++){
21             x[i][j]=0;
22             for(int k=0;k<c1;k++){
23                 x[i][j]=a[i][k]+b[k][j];
24             }
25         }
26     }
27
28 void display(int r,int c,int a[r][c]){
29     for(int i=0;i<r;i++){
30         for(int j=0;j<c;j++){
31             printf("%d ",a[i][j]);
32         }
33         printf("\n");
34     }
35 }
36
37 int main(){
38     int ch,r,c;
39
40     printf("1. Left Distributive Property\n");
41     printf("2. Right Distributive Property\n");
42     printf("3. Exit\n");
43     printf("Enter choice: ");
44     scanf("%d",&ch);
45
46     if(ch==3)
47         return 0;
48
49     printf("Enter rows and columns: ");
50     scanf("%d%d",&r,&c);
51 }
```

```
File Edit Search View Document Help
~day34.c - Mousepad

39 printf("Enter choice: ");
40 scanf("%d",&ch);
41
42 if(ch==1)
43     return 0;
44
45 printf("Enter rows and columns: ");
46 scanf("%d%d",&r,&c);
47
48 int A[r][c],B[r][c],C[r][c];
49 int X[r][c],Y[r][c],LHS[r][c],RHS[r][c];
50
51 printf("Enter Matrix A:\n");
52 read(r,c,A);
53
54 printf("Enter Matrix B:\n");
55 read(r,c,B);
56
57 printf("Enter Matrix C:\n");
58 read(r,c,C);
59
60 if(ch==1){
61     add(r,c,B,C,X);
62     mul(r,c,c,A,X,LHS);
63
64     mul(r,c,c,A,B,X);
65     mul(r,c,c,A,C,Y);
66     add(r,c,X,Y,RHS);
67 }
68 else if(ch==2){
69     add(r,c,A,B,X);
70     mul(r,c,c,X,C,LHS);
71
72     mul(r,c,c,A,C,X);
73     mul(r,c,c,B,C,Y);
74     add(r,c,X,Y,RHS);
75 }
76
77 printf("LHS:\n");
78 display(r,c,LHS);
79
80 printf("RHS:\n");
81 display(r,c,RHS);
82
83 return 0;
84 }
85
```

>OUTPUT:

```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
~ (priyadarshna@kali) ~
$ gcc day34.c -o day34
~ (priyadarshna@kali) ~
$ ./day34
1. Left Distributive Property
2. Right Distributive Property
3. Exit
Enter choice: 1
Enter rows and columns: 2 2
Enter Matrix A:
1 2
3 4
Enter Matrix B:
1 0
0 1
Enter Matrix C:
2 1
1 2
LHS:
5 7
11 15
RHS:
5 7
11 15
~ (priyadarshna@kali) ~
$ ./day34
1. Left Distributive Property
2. Right Distributive Property
3. Exit
Enter choice: 2
Enter rows and columns: 2 2
Enter Matrix A:
1 2
3 4
Enter Matrix B:
2 1
1 2
Enter Matrix C:
1 0
0 1
LHS:
3 3
4 6
RHS:
3 3
4 6
~ (priyadarshna@kali) ~
$ ./day34
1. Left Distributive Property
2. Right Distributive Property
3. Exit
Enter choice: 3
~ (priyadarshna@kali) ~
$
```